Algebra Review: Unit 7 — Polynomials & Factoring

Station 1 of 4: Adding and Subtracting Polynomials

Name: _____ Date: _____

Write each answer in standard form (highest degree first). Diagrams are not drawn to scale.

1. Write $5x - 2x^3 + 4$ in standard form. Then state its degree and name it by number of terms.

2. State the degree of $4x^4 - 9$ and classify it by number of terms.

3. Add: $3x^2 + 5x - 2 + x^2 - 8x + 6$

4. Subtract: $7x^2 - 4x + 1 - \left(2x^2 + x - 5\right)$

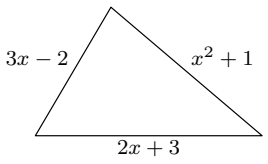
5. Add: $x^3 + 2x + 4x^3 - x^2 - 6$

6. Subtract: $5a^2 - 3a - \left(-2a^2 + a - 7\right)$

7. Add: $4x^2y + 3xy + x^2y - 5xy$

8. Subtract: $6x^3 - x + 9 - \left(6x^3 + 2x^2 - x\right)$

9. A triangle has sides $(2x + 3)$, $(x^2 + 1)$, and $(3x - 2)$. Write a polynomial for its perimeter.



10. Subtract: $9y^2 - 4y + 3 - \left(3y^2 + 2y - 8\right)$

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Station 2 of 4: Multiplying Polynomials

Name: _____ Date: _____

Multiply and write each product in standard form. Diagrams are not drawn to scale.

1. Multiply: $3x\left(4x^2-5\right)$

2. Multiply: $-2a\left(a^2+3a-4\right)$

3. Multiply: $(x+4)(x+7)$

4. Multiply: $(x-5)(x+3)$

5. Multiply: $(2x+1)(3x-4)$

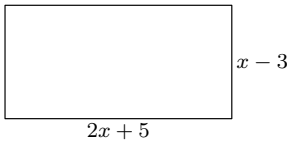
6. Multiply: $(x+6)^2$

7. Multiply: $(3x-2)^2$

8. Multiply: $(x+8)(x-8)$

9. Multiply: $(x+2)\left(x^2-3x+5\right)$

10. A rectangle has length $(2x+5)$ and width $(x-3)$. Write an expression for its area.



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Station 3 of 4: Factoring: GCF and Trinomials with $a = 1$

Name: _____ Date: _____

Factor each expression completely. Check by multiplying.

1. Factor completely: $6x + 18$

2. Factor completely: $12x^2 - 8x$

3. Factor completely: $15a^3b - 10a^2b^2$

4. Factor completely: $x^2 + 7x + 12$

5. Factor completely: $x^2 - 9x + 20$

6. Factor completely: $x^2 + 2x - 24$

7. Factor completely: $x^2 - 5x - 14$

8. Factor completely: $x^2 - 11x + 28$

9. Factor completely: $3x^2 + 21x + 36$

10. Factor completely: $x^2 + 3x - 40$

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Station 4 of 4: Factoring Special Products and Trinomials with $a \neq 1$

Name: _____ Date: _____

Factor each expression completely. Look for a GCF first.

1. Factor completely: $x^2 - 49$

2. Factor completely: $4x^2 - 25$

3. Factor completely: $x^2 + 10x + 25$

4. Factor completely: $9x^2 - 12x + 4$

5. Factor completely: $2x^2 + 7x + 3$

6. Factor completely: $3x^2 - 10x + 8$

7. Factor completely: $6x^2 + x - 12$

8. Factor completely: $5x^2 - 45$

9. Factor completely: $2x^3 - 18x$

10. Factor completely: $4x^2 + 4x - 24$

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Answer Key

Station 1: Adding and Subtracting Polynomials

1. $-2x^3 + 5x + 4$; degree 3; trinomial
2. Degree 4; binomial
3. $4x^2 - 3x + 4$
4. $5x^2 - 5x + 6$
5. $5x^3 - x^2 + 2x - 6$
6. $7a^2 - 4a + 7$
7. $5x^2y - 2xy$
8. $-2x^2 + 9$
9. $x^2 + 5x + 2$
10. $6y^2 - 6y + 11$

Station 2: Multiplying Polynomials

1. $12x^3 - 15x$
2. $-2a^3 - 6a^2 + 8a$
3. $x^2 + 11x + 28$
4. $x^2 - 2x - 15$
5. $6x^2 - 5x - 4$
6. $x^2 + 12x + 36$
7. $9x^2 - 12x + 4$
8. $x^2 - 64$
9. $x^3 - x^2 - x + 10$
10. $2x^2 - x - 15$

Station 3: Factoring: GCF and Trinomials with $a = 1$

1. $6\,(x + 3)$
2. $4x\,(3x - 2)$
3. $5a^2b\,(3a - 2b)$
4. $(x + 3)\,(x + 4)$
5. $(x - 5)\,(x - 4)$
6. $(x - 4)\,(x + 6)$
7. $(x - 7)\,(x + 2)$
8. $(x - 7)\,(x - 4)$
9. $3\,(x + 3)\,(x + 4)$
10. $(x - 5)\,(x + 8)$

Station 4: Factoring Special Products and Trinomials with $a \neq 1$

1. $(x - 7)\,(x + 7)$
2. $(2x - 5)\,(2x + 5)$
3. $(x + 5)^2$
4. $(3x - 2)^2$
5. $(x + 3)\,(2x + 1)$
6. $(x - 2)\,(3x - 4)$
7. $(2x + 3)\,(3x - 4)$
8. $5\,(x - 3)\,(x + 3)$
9. $2x\,(x - 3)\,(x + 3)$
10. $4\,(x - 2)\,(x + 3)$